Lecture 3: Numerical Differentiation

Introduction to Data Analysis Techniques

1 Taylor Approximation

1.1 Taylor Polynomial and Remainder

Continuity, differentiability, and smooth function spaces, real-analysis

We write $C^{\infty}_{\mathbb{R}\to\mathbb{R}}$ for functions $\mathbb{R} \to \mathbb{R}$ whose derivatives of every order, including order zero, are continuous

Definition 1.1. [def:taylor-polynomial-generator] The Taylor term constructor takes f, then a centre a, then an order n, and returns the nth term as a polynomial function

$$\begin{aligned}\text{TaylorGen} : C^{\infty}_{\mathbb{R}\to\mathbb{R}} &\to \mathbb{R} \to \mathbb{N} \to (\mathbb{R} \to \mathbb{R}) \\ &\equiv f \mapsto \big(a \mapsto \big(n \mapsto \big(x \mapsto \frac{\mathrm{D}^n(f)(a)}{n!} \cdot (x-a)^n\big)\big)\big)\end{aligned}$$

Definition 1.2. [def:taylor-polynomial] The n-th order Taylor polynomial generator is defined as

$$\text{TaylorPoly} : C^{\infty}_{\mathbb{R}\to\mathbb{R}} \to \mathbb{R} \to \mathbb{N} \to \mathbb{R} \to \mathbb{R} \equiv f \mapsto a \mapsto N \mapsto x \mapsto \sum^{N} n \mapsto \text{TaylorGen}(f)(a)(n)(x)$$

Convention 1.1. [conv:taylor-polynomial-notation]

Definition 1.3. [def:taylor-remainder]

Theorem 1.1 (Finite Taylor Expansion)**.** [thm:finite-taylor-expansion] *Let $f \in C^{\infty}_{\mathbb{R}\to\mathbb{R}}$, $n \in \mathbb{N}$, and $a \in I$, where $I \subseteq \mathbb{R}$ is an open interval. For the Taylor polynomial,*

$$\begin{aligned}&\forall x \in I \setminus \{a\}, \quad \exists \xi \in \big(\min(a,x), \max(a,x)\big) : \\ &f(x) = P_{f,a,n}(x) + \frac{\mathrm{D}^{n+1}(f)(\xi)}{(n+1)!} \cdot (x-a)^{n+1}\end{aligned}$$

At $x = a$, $f(a) = P_{f,a,n}(a)$ and the remainder is zero

1.2 Taylor Series

Definition 1.4 (Taylor Series)**.** [def:taylor-series] Let $f \in C^{\infty}_{\mathbb{R}\to\mathbb{R}}$ and $a \in \mathbb{R}$. Its *Taylor-series sum* at x, when it exists, is

$$f(a) + \sum \big(n \mapsto \text{TaylorGen}(f)(a)(n)(x)\big)$$

The sequence is indexed by $\mathbb{N}_1$. The displayed expression equals $f(x)$ precisely when $R_{f,a,N}(x) \to 0$ as $N \to \infty$

Remark 1.1 (Taylor Series Approximation)*.* [rem:taylor-series-approx] We can truncate the Taylor series to obtain a finite approximation. The Taylor expansion with remainder expresses its error under suitable differentiability assumptions

Adding terms does not automatically improve the approximation. The remainder must be controlled; equality with the infinite series requires that the remainder tend to zero as the number of terms increases

1.3 Example: Electric Field of a Dipole

Example 1.1 (Electric Field of a Dipole for $x \gg a$)**.** [eg:dipole-field] Consider the electric field on the axis of a dipole with charges $+q$ at $x = a$ and $-q$ at $x = -a$, where $a > 0$ and $x > a$

$$E = \frac{q}{4\pi\varepsilon_0} \cdot \left[\frac{1}{(x-a)^2} - \frac{1}{(x+a)^2} \right]$$

Expanding in a/x using the Taylor series gives

$$E = \frac{q}{4\pi\varepsilon_0 \cdot x^2} \cdot \left[4\frac{a}{x} + 8\left(\frac{a}{x}\right)^3 + 12\left(\frac{a}{x}\right)^5 + \cdots \right]$$

Keeping the first two nonzero terms gives an approximation for $x \gg a$

$$E \approx \frac{q}{4\pi\varepsilon_0 \cdot x^2} \cdot \left[4\frac{a}{x} + 8\left(\frac{a}{x}\right)^3 \right]$$

The leading term is proportional to x^{-3}

1.4 Taylor Expansion in Two Variables

TODO: Write the expansion through second order, including the mixed derivative

2 Error Orders

TODO: Define big-O notation as the step size tends to zero; explain how halving the step changes an error of order one or two

3 Numerical Differentiation

3.1 Forward and Backward Differences

TODO: Define both first-derivative approximations and derive their error orders from Taylor expansion

3.2 Central Differences

TODO: Derive central approximations for the first and second derivatives and their error orders

3.3 Programming Exercise

TODO: Implement the difference formulas for sine at x=1; compare with the exact derivatives over several step sizes

4 First-Order Differential Equations

TODO: Define an ordinary differential equation and an initial-value problem in separate blocks

5 Euler's Method

TODO: Introduce the time grid and numerical iterates; derive the forward Euler update and distinguish local and global errors

5.1 Programming Exercise

TODO: Solve x'=-x with x(0)=1 and compare with exp(-t) using several time steps

6 Second-Order Dynamics

TODO: Introduce velocity to rewrite a second-order equation as two first-order equations; derive Euler updates using the old values on both right-hand sides

6.1 Programming Exercise

TODO: Implement the pendulum model and compare results as the time step changes